MILITARY SPECIFICATION

CONVERTERS, ELECTRONIC FREQUENCY, CV-1159A/WLR-1( ), CV-1160A/WLR-1( ), CV-1161A/WLR-1( )

This specification is approved for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers the modification to convert the CV-1159/WLR-1, CV-1160/WLR-1 and CV-1161/WLR-1 Electronic Frequency Converters to solid-state converter units. The converters shall be solid-state, analog controlled, superheterodyne converter units that operate over the following frequency ranges:

TYPE 1	CV-1159A/WLR-1( )	(1.010 - 2.600) GHz)
TYPE 2	CV-1160A/WLR-1( )	(2.575 - 4.450) GHz)
TYPE 3	CV-1161A/WLR-1( )	(4.406 - 7.375) GHz)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

- MIL-S-901 - Shock Tests, H.I. (High Impact), Shipboard Machinery, Equipment and Systems; Requirements for
- MIL-E-16400 - Electronic Equipment, Naval Ship and Shore, General Specification
- MIL-E-17555 - Electronic and Electrical Equipment, Accessories, and Repair Parts; Preparation for Delivery for

STANDARDS

MILITARY

- MIL-STD-461 - Electronic Interference Characteristics; Requirements for
- MIL-STD-721 - Definitions of Effectiveness Terms for Reliability, Maintainability, Human Factors, and Safety
- MIL-STD-781 - Reliability Tests; Exponential Distribution
- MS-17325 - Meter, Time Totalizing, Digital, 115 Volt, 60 or 400 Hertz

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PUBLICATIONS

MILITARY

NAVSHIPS 0967-037-2310 Supplementary Manual for AN/WLR-1( ) with Electronic Frequency Converters CV-1159A, CV-1160A, CV-1161A.

DRAWINGS

INDUSTRY

P325006001	- Resistors, Variable, Non-Wirewound, Precision
663060620	- Regulator Board
663070220	- Assembly Parts, RF
663080220	- Cable Assembly Parts, Transformer; Power, Step-Down Parts, and Connectors, Coaxial, RF
663090209	- Mixer, Preamplifier Assembly
P85200601	- Gear Head

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General requirements. The electronic frequency control converters shall conform to the general requirements of MIL-E-16400, except as otherwise specified herein. The Electronic Frequency Converter units shall be modified in accordance with the drawings as follows:

Converter Unit	Reference Designation	Drawing Number
Type 1	CV-1159A/WLR-1( )	663060620
Type 2	CV-1160A/WLR-1( )	663070220
Type 3	CV-1161A/WLR-1( )	663080220

3.1.1 First article sample. Prior to beginning production, one sample of each style shall be tested as specified in 4.2.1 (see 6.3).

3.1.2 Transient voltage. The equipment shall conform to the transient voltage requirements as specified in MIL-E-16400.

3.1.3 Elapsed time meter. An elapsed time meter (10,000 hours minimum), not resettable to zero, is required to indicate the amount of time that each converter unit has been in operation. The elapsed time meter shall conform to the requirements of MS-17325 and shall be internally mounted.

3.1.4 Standard parts. All parts used in the converter units shall conform to the following requirements:

(a) Regulator board assembly parts. Drawing numbers 663060620, 663070220 and 663080220.
(b) RF cable assembly parts. Drawing numbers 663060620, 663070220 and 663080220.

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(c) Transformer; power, step-down. Drawing numbers 663060620, 663070220 and 663080220.
(d) Resistors; variable, non-wirewound, precision. Drawing number P325006001.
(e) Connectors; coaxial, RF. Drawing numbers 663060620, 663070220 and 663080220.
(f) Gear head. Drawing number P85200601.
(g) Mixer; preamplifier assembly. Drawing number K663090209.

3.1.4.1 Parts, other than those specified in 3.1.4 shall be selected in accordance with MIL-E-16400.

3.1.5 Equipment enclosure requirements. The chassis of each modified converter type shall be capable of being reinstalled in the original enclosure (cabinet).

3.1.6 Environmental requirements.

3.1.6.1 Temperature. The frequency converter shall conform to the Class 4 temperature requirements of MIL-E-16400.

3.1.6.2 Shock. The frequency converter shall conform to the shock requirement of MIL-E-16400 and Type A, Class 2 of MIL-S-901.

3.1.7 Electromagnetic interference. The frequency converter shall conform to the requirements of Class IA of MIL-STD-461.

3.2 Reliability. The specified mean-time-between-failures (MTBF) of the frequency converter shall be a minimum of 2,000 hours.

3.3 Design requirements.

3.3.1 Tuning inputs. The frequency converters shall be capable of being tuned by an analog function generated in the Control Storer C-2697( )/WLR-1 described in NAVSHIPS 0967-037-2310.

3.3.2 Frequency converter functional operation. The functional operation of the frequency converter units shall be as specified in drawing numbers 663060620, 663070220 and 663080220.

3.3.2.1 Modes of operation. The frequency converter units shall be capable of both manual and automatic frequency tuning. Control Storer C-2697( )/WLR-1( ) shall provide the tuning control.

3.3.2.2 Tuning range. The tuning range of the frequency converter units shall be as specified herein.

3.3.2.2.1 Frequency coverage. The minimum frequency tuning range provided by each frequency converter type, as measured at the center frequency, f_o, of the converter RF passband shall be:

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Converter Unit	Reference Designation	f Low	f High
Type 1	CV-1159A/WLR-1( )	1.010 GHz	2.600 GHz
Type 2	CV-1160A/WLR-1( )	2.575 GHz	4.450 GHz
Type 3	CV-1161A/WLR-1( )	4.406 GHz	7.375 GHz

3.3.2.2.2 Noise figure. The maximum noise figure of each frequency converter type as measured from the converter RF (Radio Frequency) input to the IF (Intermediate Frequency) output shall not exceed 18 dB (maximum) over the entire frequency bands specified in 3.3.2.2.1. Ripple shall not exceed plus or minus 2 dB as measured within a 20 MHz bandwidth at the 1.5 dB response points for each converter type.

3.3.2.2.3 Intermediate frequency. The intermediate frequencies of the frequency converter units shall be 160 MHz ±0.1 percent and 400 MHz ±0.1 percent.

3.3.2.2.4 RF bandwidth. The minimum RF bandwidth of each frequency converter type as measured from the IF center frequency, f_o, at the IF output port shall be ±10 MHz minimum at the 1.5 dB response points. Ripple shall not exceed ±1 dB.

3.3.2.2.5 Spurious responses. Spurious responses as referenced to the RF input port of each frequency converter unit shall not exceed -75 dB as referenced to 1 milliwatt (dBm).

3.3.2.2.6 Dynamic range. The 1 dB compression point of the frequency converter unit IF output shall not occur at an input power level lower than -30 dBm.

3.3.2.2.7 Image rejection. The image rejection of the frequency converter shall be 65 dB (minimum).

3.3.2.2.8 Tuner design. The Local Oscillator (LO) shall be a mechanical tuned transferred electron device and shall conform to the requirements specified in the drawings indicated in 3.1.

3.3.2.2.8.1 Local oscillator radiation. The level of any internally generated signal shall not exceed -35 dBm as measured at the RF input port.

3.3.2.2.8.2 Local oscillator accuracy. The LO accuracy shall be ±0.2 percent (minimum) over the entire frequency bands listed in 3.3.2.2.1 at 20° to 30° centigrade (C) and ±1 percent at temperature extremes.

3.3.2.2.8.3 Local oscillator frequency coverage. The minimum frequency tuning range of the local oscillator provided by each frequency converter type as measured at the center frequency (f_o) of the converter RF passband shall be:

Converter Unit	Reference Designation	Low	High
Type 1	CV-1159A/WLR-1( )	1.410 GHz	3.000 GHz
Type 2	CV-1160A/WLR-1( )	2.175 GHz	4.054 GHz
Type 3	CV-1161A/WLR-1( )	4.806 GHz	7.775 GHz

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3.3.2.2.8.4 Local oscillator resetability. The local oscillator shall be capable of being reset to within ±0.10 percent of any frequency in the indicated frequency bands of 3.3.2.2.8.3.

3.3.2.2.8.5 Sweep rate. The frequency converter unit shall scan the entire frequency bands of 3.3.2.2.1 within 2.4 seconds.

3.3.2.2.8.6 RF input impedence. The RF input impedence of the frequency converter shall be 50 ohms (nominal).

3.3.2.2.8.7 AM noise. The local oscillator AM double side band (DSB) noise level shall not exceed -83 dBm/DSB in a 20 MHz bandwidth 400 MHz from the carrier.

3.3.2.2.8.8 Mixer-preamplifier. The mixer-preamplifier shall be a combined integral solid-state package. The minimum conversion gain shall be 5 dB with a maximum noise figure of 12 dB from 1.010 GHz to 7.375 GHz.

3.3.2.2.8.9 RF burnout protection. The components of the frequency converter units shall not be burned out after subjection to an in-band or out-of-band 700 milliwatt (mW) continuous wave (CW) signal in each of the respective bands of 3.3.2.2.1.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 General inspection. The general inspection requirements shall be in accordance with MIL-E-16400 and shall consist of the first article inspection of 4.2.1, the production inspection of 4.2.2.1 and production control inspection of 4.2.2.2.

4.2.1 First article inspection. Unless otherwise specified (see 6.3), three units (one unit from each style) shall be required for first article inspection. First article inspection shall consist of all examination and testing necessary to determine compliance with the requirements of this specification. First article inspection shall be in accordance with the preproduction inspection of MIL-E-16400, except the water cooling, enclosure, accelerated life, inclination, noise and magnetic tests will not be required.

4.2.1.1 Operating test. The operating test of MIL-E-16400 shall include tests to determine conformance with the following requirements:

(a) Frequency coverage (see 3.3.2.2.1)
(b) Noise figure (see 3.3.2.2.2)

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(c) RF bandwidth (see 3.3.2.2.4)
(d) Spurious responses (see 3.3.2.2.5)
(e) Dynamic range (see 3.3.2.2.6)
(f) Image rejection (see 3.3.2.2.7)
(g) LO radiation (see 3.3.2.2.8.1)
(h) LO accuracy (see 3.3.2.2.8.2)
(i) LO resetability (see 3.3.2.2.8.4)
(j) Sweep rate (see 3.3.2.2.8.5)
(k) RF input impedence (see 3.3.2.2.8.6)
(l) AM noise (see 3.3.2.2.8.7)
(m) Conversion gain and noise figure (see 3.3.2.2.8.8)
(n) RF burnout protection (see 3.3.2.2.8.9)

4.2.1.2 Burn-in-test. Each equipment submitted for pre-production inspection shall have passed 48 hours of continuous failure free burn-in with operating tests (4.2.1.1) being performed before and after the burn-in test. The failures which occur and the performance parameters measured and their values shall be recorded each time a unit is submitted for burn-in. Any degraded performance shall be cause for stopping the test, repairing the equipment, and restarting the test at zero time.

4.2.1.3 Reliability demonstration. A reliability demonstration test (Qualification) shall be conducted on the frequency converter units to evaluate the design limitations in terms of mean-time-between-failures. The test shall be conducted in accordance with MIL-STD-781. Test Plan IVA, Test Level A of MIL-STD-781 shall be used for this demonstration. The three (one from each type) first article models shall be used for this demonstration. In the event a reject decision is reached, a retest shall be conducted following the incorporation of appropriate design, part, or production process changes. The retest shall be conducted until such time that an accept decision using Test Plan IVA, Test Level A, has been reached or direction to terminate is received from the procuring activity.

4.2.2 Quality conformance inspection.

4.2.2.1 Production inspection. Production inspection shall be conducted on each equipment in accordance with MIL-E-16400. The operating test shall be in accordance with 4.2.2.1.1.

4.2.2.1.1 Operating test. The operating test shall include tests to determine conformance with the following requirements: (Swept frequency testing required.)

(a) Noise figure (see 3.3.2.2.2)
(b) Frequency coverage (see 3.3.2.2.1)
(c) LO accuracy (see 3.3.2.2.8.2)
(d) RF Burn-out protection (see 3.3.2.2.8.9)

4.2.2.1.2 Burn-in test. Each equipment submitted for production inspection shall have passed 48 hours of continuous, failure free, burn-in with an operating test per 4.2.2.1.1 being performed before and after the burn-in test. The failures which occur and the performance parameters measured and their

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values shall be recorded each time a unit is submitted for burn-in. Any degraded performance shall be cause for stopping the test, repairing the equipment and restarting the test at zero time.

4.2.2.2 Production control inspection. Production control inspection shall be conducted on one equipment out of each lot of 75 or part thereof and shall consist of the following tests of MIL-E-16400:

(a) Heat
(b) Temperature
(c) Power
(d) Controls and control circuits
(e) Reliability (see 4.2.2.2.1)
(f) Vibration (see 4.3)

4.2.2.2.1 Reliability test. This demonstration test shall be a fixed length test of 200 hours with no failures. Reliability tests shall be performed in accordance with 4.2.1.3.

4.3 Vibration test. For the vibration test of MIL-E-16400, the test sample shall be mounted in the normal operating condition. The sample shall be operating before, during, and at the completion of the vibration test.

4.4 Failures. In addition to the definition of a failure as given in MIL-STD-721, and any definition given in the applicable reliability test specification or standard, the following criteria for the determination of a failure shall apply:

(a) Whenever any of the performance characteristics are outside of the requirements of the specifications at any specified environmental conditions, at least one failure has occurred.
(b) An analysis of the cause of each failure shall be made in accordance with the applicable reliability test specification or standard.

4.4.1 Failure classifications. The failure categories of MIL-STD-781 shall apply. No failure may be classified non-relevant or reclassified non-relevant without written approval of the procuring activity. The following failure categories shall be in addition to those of MIL-STD-781:

(a) Relevant failures.

(1) Part failures. Failures due to defective parts shall be classified as relevant failures. In the event that several parts of the same type fail during the test, each one shall be considered a separate relevant failure, unless it can be shown that one failure caused one or more of the others (see MIL-STD-781, dependent failures).
(2) Wearout parts. Certain parts of known limited life, such as batteries, may have a life stipulated prior to the initiation of testing as approved by the procuring activity. Failures of these parts occurring prior to the end of the stipulated period are relevant. Failures

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of these parts occurring after the stipulated period are nonrelevant, but any dependent failures caused thereby are relevant.

(3) Multiple failures. In the event simultaneous part failures occur, each failed part which would independently prevent satisfactory equipment performance shall be counted as a relevant failure except as follows: If the contractor and the procuring activity agree that the failure of one part was entirely responsible for the failure of any other parts, then each such dependent part failure shall not be counted as a relevant failure. At least one equipment relevant failure shall be counted when a dependent failure is claimed.

(4) Intermittent failures. The first occurrence of an intermittent failure on any one equipment shall be counted as a relevant failure, although subsequent occurrences of the same intermittency on that same unit will be considered nonrelevant. However, no equipment shall be shipped for which an intermittent failure remains unresolved without specific approval of the procuring activity.

(5) Adjustments.

(a) Accessible controls. Each adjustment of a control which is accessible to the operator during normal use is a relevant failure if the information necessary to restore equipment operation is not available from indicators which are an integral part of the equipment under test.

(b) Inaccessible controls. Each adjustment of a control which is inaccessible to the operator during normal use is a relevant failure, unless a period of operating (not standby) time has elapsed since the last adjustment of any inaccessible control on that equipment, which is equal to or greater than the period specified for the equipment operational stability.

(6) Failures of built-in test features. Any malfunction (including a false alarm) of the built-in test features of the equipment shall be classified as a relevant failure.

(b) Nonrelevant failures. Although nonrelevant failures are not used for MTBF calculations, all failures shall be recorded and reported. Only those failures listed hereinafter may be counted as nonrelevant.

(1) Failures directly attributable to improper installation in the test chamber.

(2) Failures of test instrumentation or monitoring equipment (other than the built-in test function).

(3) Failures resulting from test operator error in setting up, or in testing the equipment.

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(4) Dependent failures, unless caused by degradation of items of known limited life. (At least one relevant failure shall be counted when a dependent failure is claimed.)
(5) Failures attributable to an error in the test procedures.
(6) The second (and any subsequent) occurrences of the same intermittent failure on the same unit.
(7) Failures occurring during burn-in, troubleshooting, repair verification, or set-up time.
(8) Malfunctions of the Time Totalizing Meters or certain lighting circuit failures, when the approved test procedures specifically designate them as non-relevant.
(9) Failures clearly attributable to an overstress condition in excess of the design requirements.
(10) Adjustments:

a. Accessible controls. Adjustments of controls which are accessible to the operator during normal use shall be counted as non-relevant failures provided the adjustment is accomplished without reference to test equipment, meters, or indicators, which are not a functional part of the equipment under test.
b. Inaccessible controls. Adjustments of controls which are not accessible to the operator during normal use are non-relevant failures provided no such adjustment has been made to the unit specified for the subsystem operational stability.

(11) Other, as explicitly defined in the approved test procedures.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements or referenced documents listed in section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation, packaging, packing and marking. The equipment, accessories, technical publications, and repair parts shall be preserved, packaged, packed and marked as specified in the contract or order in accordance with MIL-E-17555 (see 6.2.1).

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6. NOTES

6.1 Intended use. These converter units will be used with AN/WLR-1 receivers installed aboard surface ships and submarines.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification
(b) Number of first article samples to be submitted if other than specified in 4.2.1.3
(c) Marking, levels of preservation and packaging and packing required (see 5.1)

6.3 First article. Invitations for bids should provide that the Government reserves the right to waive the requirement for first article samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

Preparing activity:
Navy - EC

(Project 5820-N640)

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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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NAVAL ELECTRONIC SYSTEMS COMMAND
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